Section 3.2 - Rates of Change

Example 1: You decide to go on a trip along Highway 6. Your destination is 142 miles away, and it takes you exactly 2 hours to get there.

a) What was your average velocity for the trip?

(0,0) (2,142)

Avg vel = $\frac{142-0}{2-0}$ = 71 miles/hour

b) After you were on the road for an hour, you pass a cop with a radar gun. Assuming the speed limit was 70 mph, were you speeding?

We don't know. We need more information.

The table below represents the car's position every hour:

x	0	1	2
y	0	90	142

c) Based on the table above, what was the average velocity for the first hour of the trip?

(0,0) (1,90)

Avg vel = $\frac{90-0}{1-0}$ = 90 miles/hour

d) What was the average velocity for the second hour of the trip?

(1,90) (2,142)

Avg vel = $\frac{142-90}{2-1}$ = 52 miles/hour

e) Use the table below to estimate the instantaneous velocity when you passed the cop. The table below represents the car's position near the one hour mark.

x	0.99	0.999	1	1.001	1.01
y	89.318	89.932	90	90.068	90.678

(.99, 1)

$\frac{90-89.318}{1-.99}$ = 68.2 mph

(1, 1.01)

$\frac{90.678-90}{1.01-1}$ = 67.8 mph

(.999, 1)

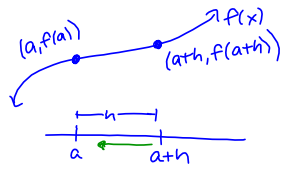
$\frac{90-89.932}{1-.999}$ = 68 mph

(1, 1.001)

$\frac{90.068-90}{1.001-1}$ = 68 mph

It appears you were going approximately 68 mph at the instant you passed the cop and therefore were not speeding

Definition: The **average velocity** of an object on the interval $t = a$ to $t = a + h$ is given by



$$\text{Avg. vel} = \frac{f(a+h)-f(a)}{\cancel{a}+h-\cancel{a}}$$

$$\boxed{\text{Avg. vel} = \frac{f(a+h)-f(a)}{h}}$$

where $y = f(t)$ gives the position of the object at time t.

Definition: The **instantaneous velocity** of an object at time $t = a$ is given by

$$\boxed{\text{Inst. vel} = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}}$$

Example 2: If a rock is thrown upward on the planet Mars with a velocity of 10 m/s, its height in meters t seconds later is given by $y = \underbrace{10t - 1.86t^2}_{f(t)}$.

a) Find the average velocity over the given time intervals:

i) $[1, 1.01]$

$$\frac{f(1.01)-f(1)}{1.01-1} = \frac{8.202614-8.14}{.01} = \boxed{6.2614 \text{ m/s}}$$

ii) $[1, 1.001]$

$$\frac{f(1.001)-f(1)}{1.001-1} = \frac{8.14627814-8.14}{.001} = \boxed{6.27814 \text{ m/s}}$$

b) Use part a) to estimate the instantaneous velocity when $t = 1$.

$$\approx \boxed{6.27814 \text{ m/s}}$$

c) Find the exact value of the instantaneous velocity when $t = 1$.

Click A on your iclicker

$$\text{Inst. vel} = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h}$$

$$= \lim_{h\to 0} \frac{\cancel{8.14}+6.28h-1.86h^2-\cancel{8.14}}{h}$$

$$= \lim_{h\to 0} \frac{6.28h-1.86h^2}{h}$$

$$= \lim_{h\to 0} \frac{\cancel{h}(6.28-1.86h)}{\cancel{h}}$$

$$= \lim_{h\to 0} (6.28-1.86h) = 6.28-1.86(0) = \boxed{6.28 \text{ m/s}}$$

$$f(1+h) = 10(1+h) - 1.86\underbrace{(1+h)^2}_{(1+h)(1+h)}$$

$$= 10+10h-1.86(1+2h+h^2)$$

$$= 10+10h-1.86-3.72h-1.86h^2$$

$$= 8.14+6.28h-1.86h^2$$

Definition: For $y = f(x)$, the **average rate of change from $x = a$ to $x = a + h$** is

* slope of the secant line

$$\boxed{AROC} = \frac{f(a+h)-f(a)}{\cancel{a}+h-\cancel{a}} = \boxed{\frac{f(a+h)-f(a)}{h}}$$

Definition: For $y = f(x)$, the **instantaneous rate of change at $x = a$** is

* slope of the tangent line

$$\boxed{IROC = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}}$$

Example 3: Estimate (and then find exactly) the instantaneous rate of change of $f(x) = x^2$ at $x = 1$.

Estimate:

$[1, 1.01]$

$$AROC = \frac{f(1.01)-f(1)}{1.01-1} = \frac{(1.01)^2-1^2}{.01} = 2.01$$

$[.99, 1]$

$$AROC = \frac{f(1)-f(.99)}{1-.99} = \frac{1^2-(.99)^2}{.01} = 1.99$$

It appears that IROC is $\boxed{\text{approx } 2}$ at $x=1$.

Exactly:

$$IROC = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h} = \lim_{h\to 0} \frac{\cancel{1}+2h+h^2-\cancel{1}}{h} = \lim_{h\to 0} \frac{2h+h^2}{h} = \lim_{h\to 0} \frac{\cancel{h}(2+h)}{\cancel{h}} = \lim_{h\to 0}(2+h) = 2+0 = \boxed{2}$$

$f(1+h)$
$= (1+h)^2$
$= (1+h)(1+h)$
$= 1+2h+h^2$

Example 4: Find the equation of the tangent line of $f(x) = \sqrt{5+x}$ at $x = 20$.

IROC at $x=20$ is the slope of the tangent line at $x=20$

$$\text{Slope of tangent line} = IROC = \lim_{h\to 0} \frac{f(20+h)-f(20)}{h} = \lim_{h\to 0} \frac{\sqrt{5+20+h}-\sqrt{5+20}}{h} = \lim_{h\to 0} \frac{\sqrt{25+h}-5}{h} \cdot \frac{(\sqrt{25+h}+5)}{(\sqrt{25+h}+5)}$$

$$= \lim_{h\to 0} \frac{\cancel{h}}{\cancel{h}(\sqrt{25+h}+5)} = \lim_{h\to 0} \frac{1}{\sqrt{25+h}+5} = \frac{1}{\sqrt{25+0}+5} = \frac{1}{10}$$

$(\sqrt{25+h}-5)(\sqrt{25+h}+5)$
$= \cancel{25}+h+\cancel{5\sqrt{25+h}}-\cancel{5\sqrt{25+h}}-\cancel{25}$
$= h$

$f(20) = \sqrt{5+20} = 5$

the slope of the tangent line

The tangent line passes through the point $(20, f(20))$

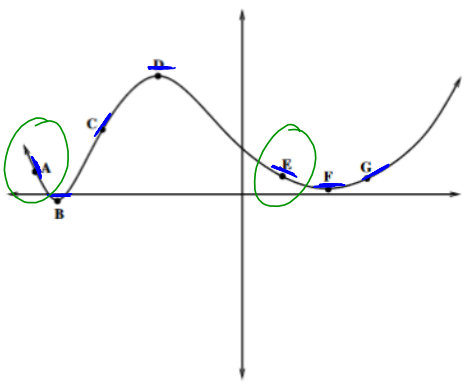
$(20, 5)$ $m = \frac{1}{10}$

$y = mx+b$
$5 = \frac{1}{10}(20)+b$
$5 = 2+b$
$3 = b$

$$\boxed{y = \frac{1}{10}x+3}$$

Example 5: For the function shown below, at what labeled points is the instantaneous rate of change

slope of the tangent line



a) positive?

C & G

b) negative?

A & E

c) zero?

B, D, & F

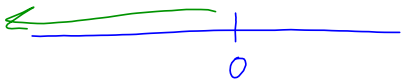
d) most negative?

$\boxed{A}$

Clicker question:

At what labeled point is the instantaneous rate of change most negative?

(a) A
(b) B
(c) C
(d) D
(e) E



Section 3.2 Highly Suggested Homework Problems: 1-43(odd), 51, 57, 59, 65, 67, 69